

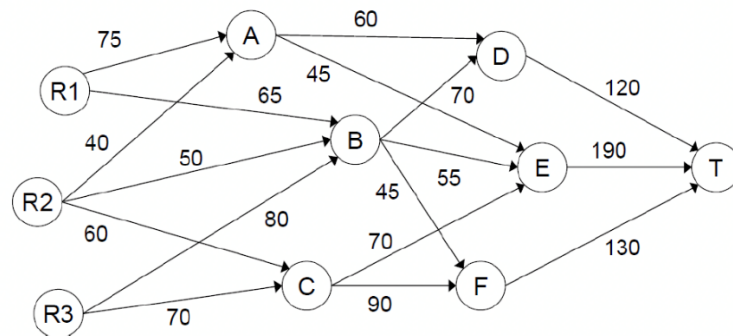
LW3 Tutorial: Network models

Task 1

A growing town relies on water supplied from three rivers: R_1 , R_2 , R_3 . These rivers are connected to the town T through a system of aqueducts. Between the rivers and the town are several junction points where flows can split or combine. Each aqueduct has a maximum daily pumping capacity, determined by pipe size, pump power, and terrain. These capacities are shown on the network diagram.

The town's water manager must decide: How much water should be sent through each aqueduct so that the total amount of water reaching the town is as large as possible, without exceeding any pipe's capacity? Develop the algebraic and Excel spreadsheet models to solve the problem and provide solution to the manager.

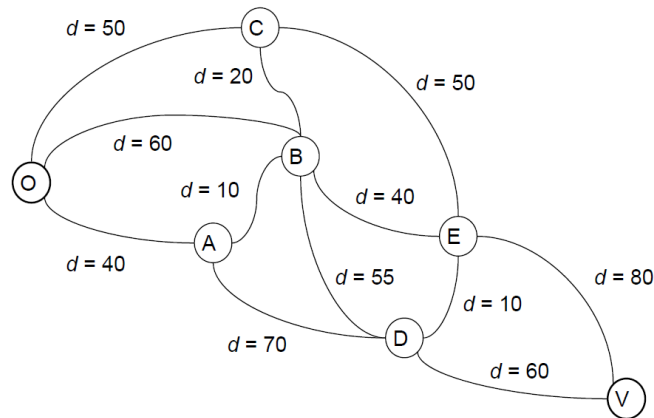
Graph depicting the water distribution network.



Task 2

A person needs to drive from an origin (**node O**) to a destination (**node V**) and wants to determine the shortest possible route. Depending on the chosen path, the trip may pass through any of five intermediate towns: A, B, C, D, and E. The mileage for each road segment that directly connects two towns (with no intermediate stops) is provided in the graph below.

Although this type of problem can be solved using heuristic algorithms such as Dijkstra's algorithm or the A* search algorithm, it can also be formulated as a network flow problem and solved using the Simplex method for linear programming. Formulate this routing problem as a network flow model and develop appropriate spreadsheet-based model to solve it.



Task 3. In a certain organisation's campus, several buildings required to be connected through fibre-optic links between selected pairs of buildings. However, budget constraints prevented the installation of every possible link. The administrator therefore imposed two design requirements:

1. The installed fibre links must be sufficient to ensure that **every building can communicate with every other building**, either directly or indirectly, through the network of installed connections.
2. The **total cost of installation**, represented by the fibre-optic cable length required for each potential connection, must be **as low as possible**.

A schematic layout of the campus is provided, in which each building is shown together with the **length of fibre** (one section of line represent 100 meters) required to connect it to neighbouring buildings. For example, from building A to B the length of fibre is 400 meters. Your task is to determine the optimal set of links that connects all buildings while minimising the total fibre length required by an efficient method.

